



Lecture 14: Bernoulli process and Poisson process

Yuan Shen

shenyuan_ee@tsinghua.edu.cn

Tsinghua University

June 1, 2021

- Date: June 16, 2021
- Time: 8:00-10:00 (GMT+08:00, Beijing time)
- Location: Lecture hall, Department Building of School of Architecture (建馆报告厅)
- Coverage: Lecture 1 to 14
- Additional information:
 - Closed-book exam, with one piece of A4 “cheat sheet” available (Two-sided)
 - Calculator **NOT** necessary
- Account for 40% in final grade
- Please get well-prepared! Good luck!
- If you cannot take the exam offline, or you have a time conflict with other exams, please fill the survey on <https://www.wjx.cn/vj/mBDIvw4.aspx> before **Jun. 6!**

1. Review
2. The Bernoulli process
3. The Poisson process
4. Recitation
5. Summary



1. Review
2. The Bernoulli process
3. The Poisson process
4. Recitation
5. Summary



- Central limit theorem

$$Z_n = \frac{S_n - \mathbf{E}[S_n]}{\sigma_{S_n}} = \frac{S_n - n\mathbf{E}[X]}{\sqrt{n}\sigma}$$

$$\mathbf{P}(Z_n \leq c) \rightarrow \mathbf{P}(Z \leq c), \quad \text{for all } c$$

- The strong law of large numbers

$$\mathbf{P}\left(\lim_{n \rightarrow \infty} \frac{X_1 + \cdots + X_n}{n} = \mu\right) = 1$$

1. Review
2. The Bernoulli process
3. The Poisson process
4. Recitation
5. Summary



- A **stochastic process** is a mathematical model of a probabilistic experiment that evolves in time and generates a sequence of numerical values.
- First view: sequence of RVs $X_1, X_2, \dots$
 - $\mathbf{E}[X_t] =$
 - $\text{var}(X_t) =$

- A **stochastic process** is a mathematical model of a probabilistic experiment that evolves in time and generates a sequence of numerical values.
- First view: sequence of RVs $X_1, X_2, \dots$
 - $\mathbf{E}[X_t] =$
 - $\text{var}(X_t) =$
- Second view: what is the right sample space?
 - $\mathbf{P}(X_t = 1 \text{ for all } t)$

- A **stochastic process** is a mathematical model of a probabilistic experiment that evolves in time and generates a sequence of numerical values.
- First view: sequence of RVs $X_1, X_2, \dots$
 - $\mathbf{E}[X_t] =$
 - $\text{var}(X_t) =$
- Second view: what is the right sample space?
 - $\mathbf{P}(X_t = 1 \text{ for all } t)$
- Examples of classic arrival-type processes
 - Bernoulli process (memoryless, discrete time)
 - Poisson process (memoryless, continuous time)
- More complicated stochastic processes next semester

Definition of the Bernoulli process

Bernoulli process is a sequence $X_1, X_2, \dots$ of independent Bernoulli RVs X_i with

$$\mathbf{P}(\text{success}) = \mathbf{P}(X_i = 1) = p$$

$$\mathbf{P}(\text{failure}) = \mathbf{P}(X_i = 0) = 1 - p$$

for each i .

- Visualization: A sequence of Bernoulli trials
 - Sequence of lottery wins/losses
 - Sequence of ups and downs of the Dow Jones
 - Arrivals (each second) to a bank
 - Arrivals (at each time slot) to server

Independence property

- For any given time n , the sequence of $X_{n+1}, X_{n+2}, \dots$ is also a Bernoulli process, and is independent from $X_1, \dots, X_n$.

Examples:

- Let U be the number of successes in trials 1 to 5. Let V be the number of successes in trials 6 to 10. Then, U and V are independent.
- Let U (respectively, V) be the first odd (respectively, even) time i in which we have a success. Then, U is determined by the odd-time sequence $X_1, X_3, \dots$, whereas V is determined by the even-time sequence $X_2, X_4, \dots$. And U and V are independent.

Memorylessness property

- Let n be a given time and let $\bar{T}$ be the time of the first success after time n . Then $\bar{T} - n$ has a geometric distribution with parameter p , and is independent of the RVs $X_1, \dots, X_n$.
- Since the future of the process is independent of the past, the number of future trials until the first success is described by

$$\mathbf{P}(\bar{T} - n = t \mid \bar{T} > n) = (1 - p)^{t-1}p = \mathbf{P}(T = t), \quad t = 1, 2, \dots$$

- Let N be the first i for which $X_{i-1} = X_i = 1$. What is the probability $\mathbf{P}(X_{N+1} = X_{N+2} = 0)$ that there are no successes in the two trials that follow?

- Let N be the first i for which $X_{i-1} = X_i = 1$. What is the probability $\mathbf{P}(X_{N+1} = X_{N+2} = 0)$ that there are no successes in the two trials that follow?

Solution:

- By conditioning on the possible values of N , we have

$$\mathbf{P}(X_{N+1} = X_{N+2} = 0) = \sum_{n=1}^{\infty} \mathbf{P}(N = n) \mathbf{P}(X_{n+1} = X_{n+2} = 0 \mid N = n)$$

- Let N be the first i for which $X_{i-1} = X_i = 1$. What is the probability $\mathbf{P}(X_{N+1} = X_{N+2} = 0)$ that there are no successes in the two trials that follow?

Solution:

- By conditioning on the possible values of N , we have

$$\mathbf{P}(X_{N+1} = X_{N+2} = 0) = \sum_{n=1}^{\infty} \mathbf{P}(N = n) \mathbf{P}(X_{n+1} = X_{n+2} = 0 \mid N = n)$$

- The event $N = n$ occurs if and only if $\{X_1, \dots, X_n\}$ satisfy the condition, but

$$\mathbf{P}(X_{n+1} = X_{n+2} = 0 \mid N = n) = \mathbf{P}(X_{n+1} = X_{n+2} = 0) = (1-p)^2$$

which leads to

$$\mathbf{P}(X_{N+1} = X_{N+2} = 0) = \sum_{n=1}^{\infty} \mathbf{P}(N = n) (1-p)^2 = (1-p)^2.$$

- Denote the time of the k th success as Y_k , and the k th interarrival time as T_k . That is,

$$T_1 = Y_1, \quad T_k = Y_k - Y_{k-1}, \quad k = 2, 3, \dots$$

represents the number of trials following the $(k - 1)$ th success until the next success.

- Note that

$$Y_k = T_1 + T_2 + \dots + T_k.$$

Alternative description of the Bernoulli process

- Start with a sequence of independent geometric RVs $T_1, T_2, \dots$, with common parameter p , and let these stand for the interarrival times.
- Record a success at times $T_1, T_1 + T_2$, etc.

- The time Y_k of the k th success is equal to the sum $Y_k = T_1 + T_2 + \cdots + T_k$ of k independent identically distributed geometric RVs.
- The mean and variance of Y_k

$$\mathbf{E}[Y_k] = \mathbf{E}[T_1] + \cdots + \mathbf{E}[T_k] = \frac{k}{p}$$

$$\text{var}(Y_k) = \text{var}(T_1) + \cdots + \text{var}(T_k) = \frac{k(1-p)}{p^2}$$

- The PMF of Y_k is given by (Pascal PMF of order k)

$$p_{Y_k}(t) = \binom{t-1}{k-1} p^k (1-p)^{t-k}, \quad t = k, k+1, \dots$$

- For $t \geq k$, we observe that the event $\{Y_k = t\}$ will occur if and only if both of the following two events A and B occur:
 - event A : trial t is a success;
 - event B : exactly $k - 1$ successes occur in the first $t - 1$ trials.

- For $t \geq k$, we observe that the event $\{Y_k = t\}$ will occur if and only if both of the following two events A and B occur:
 - event A : trial t is a success;
 - event B : exactly $k - 1$ successes occur in the first $t - 1$ trials.
- The probabilities of these two independent events are

$$\mathbf{P}(A) = p$$

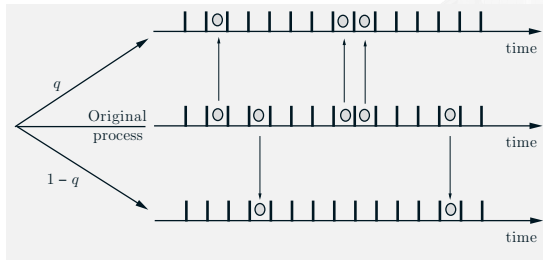
$$\mathbf{P}(B) = \binom{t-1}{k-1} p^{k-1} (1-p)^{t-k}.$$

- Therefore,

$$p_{Y_k}(t) = \mathbf{P}(Y_k = t) = \mathbf{P}(A)\mathbf{P}(B) = \binom{t-1}{k-1} p^k (1-p)^{t-k}.$$

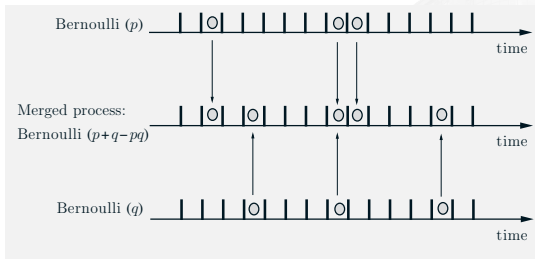
Splitting of a Bernoulli process

- Whenever there is an arrival, we choose to either keep it (with probability q), or to discard it (with probability $1 - q$).
- **Claim:** Both the process of arrivals that are kept and the process of discarded arrivals are Bernoulli processes, with success probability pq and $p(1 - q)$, respectively, at each time.



Merging of a Bernoulli process

- In a reverse situation, we start with two independent Bernoulli processes (with parameters p and q respectively). An arrival is recorded in the merged process if and only if there is an arrival in at least one of the two original processes.
- The merged process is Bernoulli, with success probability $p+q-pq$ at each time step.



1. Review
2. The Bernoulli process
- 3. The Poisson process**
4. Recitation
5. Summary



Definition of the Poisson Process

An arrival process is called a **Poisson process** with rate λ if it has the following properties:

- **Time homogeneity**

$$\mathbf{P}(k, \tau) = \mathbf{P}(k \text{ arrivals in interval of duration } \tau)$$

- **Independence**

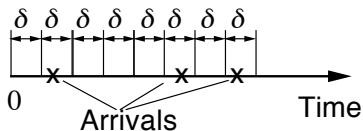
Numbers of arrivals in disjoint time intervals are **independent**.

- **Small interval probabilities**

$$\mathbf{P}(k, \tau) \approx \begin{cases} 1 - \lambda\tau + o(\tau), & \text{if } k = 0 \\ \lambda\tau + o_1(\tau), & \text{if } k = 1 \\ o_k(\tau), & \text{if } k > 1. \end{cases}$$

Bernoulli/Poisson Relation

- The relation between Bernoulli and Poisson process

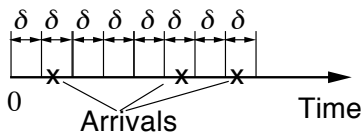


$$n = t/\delta$$

$$p = \lambda \delta$$

$$np = \lambda t$$

- The relation between Bernoulli and Poisson process



$$n = t/\delta$$

$$p = \lambda\delta$$

$$np = \lambda t$$

- For a binomial PMF $p_S(k; n, p)$, we let n tend to infinity,

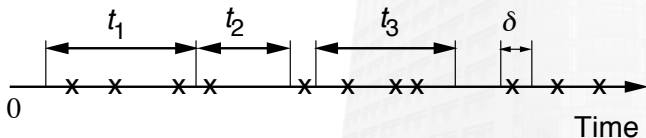
$$\begin{aligned}\lim_{n \rightarrow \infty} p_S(k; n, p) &= \lim_{n \rightarrow \infty} \frac{n!}{(n-k)!k!} p^k (1-p)^{n-k} \\ &= \lim_{n \rightarrow \infty} \frac{n(n-1) \cdots (n-k+1)}{n^k} \cdot \frac{(\lambda t)^k}{k!} \cdot \left(1 - \frac{\lambda t}{n}\right)^{n-k} \\ &= 1 \cdot \frac{(\lambda t)^k}{k!} \cdot e^{-\lambda t} = \mathbf{P}(k, t)\end{aligned}$$

PMF of Number of Arrivals N

- Finely discretize $[0, t]$: approximately Bernoulli
- N_t (of discrete approximation): binomial
- Taking $\delta = t/N_t \rightarrow 0$ (or $N_t \rightarrow \infty$) gives:

$$\mathbf{P}(k, \tau) = \frac{(\lambda\tau)^k e^{-\lambda\tau}}{k!}, \quad k = 0, 1, \dots$$

- $\mathbf{E}[N_t] = \lambda t$
- $\text{var}(N_t) = \lambda t$



Time T of the first arrival

- Time T of the first arrival

$$F_T(t) = \mathbf{P}(T \leq t) = 1 - \mathbf{P}(T > t) = 1 - \mathbf{P}(0, t) = 1 - e^{-\lambda t}, \quad t \geq 0.$$

- Differentiate the CDF to obtain the PDF formula

$$f_T(t) = \lambda e^{-\lambda t}, \quad t \geq 0$$

- T is exponentially distributed with parameter λ .
- **Memoryless property:** The time to the next arrival is independent of the past

Example

- You get email according to a Poisson process at a rate of $\lambda = 0.2$ messages per hour. You check your email every hour. What is the probability of finding 0 and 1 new messages?



- You get email according to a Poisson process at a rate of $\lambda = 0.2$ messages per hour. You check your email every hour. What is the probability of finding 0 and 1 new messages?
- These probabilities can be found using the Poisson pmf with $\tau = 1$, and $k = 0$ or $k = 1$.

$$\mathbf{P}(0, 1) = e^{-0.2} = 0.819, \quad \mathbf{P}(1, 1) = 0.2 \cdot e^{-0.2} = 0.164.$$

- You get email according to a Poisson process at a rate of $\lambda = 0.2$ messages per hour. You check your email every hour. What is the probability of finding 0 and 1 new messages?
- These probabilities can be found using the Poisson pmf with $\tau = 1$, and $k = 0$ or $k = 1$.

$$\mathbf{P}(0, 1) = e^{-0.2} = 0.819, \quad \mathbf{P}(1, 1) = 0.2 \cdot e^{-0.2} = 0.164.$$

- The probability of finding no new messages for a whole day,

$$\mathbf{P}(0, 24) = e^{-0.2 \cdot 24} = 0.0083.$$

- It can also be interpreted as finding no messages during each of 24 hours,

$$\mathbf{P}(0, 24) = (\mathbf{P}(0, 1))^{24} = (e^{-0.2})^{24} = 0.0083.$$

- We also denote the time of the k th success as Y_k , and denote the k th interarrival time as T_k . That is,

$$T_1 = Y_1, \quad T_k = Y_k - Y_{k-1}, \quad k = 2, 3, \dots$$

represents the time following the $(k-1)$ th success until the next success.

- Note that

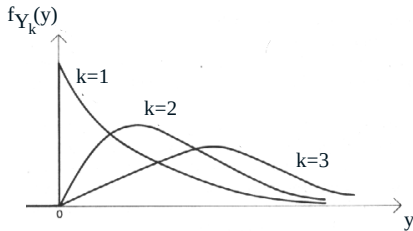
$$Y_k = T_1 + T_2 + \dots + T_k.$$

Alternative description of the Poisson process

- Start with a sequence of independent exponential RVs $T_1, T_2, \dots$, with common parameter λ , and let these stand for the interarrival times.
- Record a success at times $T_1, T_1 + T_2$, etc.

- Time of the k th arrival, Y_k : **Erlang** distribution

$$f_{Y_k}(y) = \frac{\lambda^k y^{k-1} e^{-\lambda y}}{(k-1)!}, \quad y \geq 0$$



- The k th arrival occurs between y and $y + \delta$ if and only if the following two events A and B occur:
 - event A : there is an arrival during the interval $[y, y + \delta]$;
 - event B : exactly $k - 1$ arrivals before time y .
- The probability of these two independent events are

$$\mathbf{P}(A) \approx \lambda\delta, \quad \mathbf{P}(B) = \mathbf{P}(k - 1, y) = \frac{\lambda^{k-1}y^{k-1}e^{-\lambda y}}{(k - 1)!}.$$

- Therefore,

$$\delta f_{Y_k}(y) \approx \mathbf{P}(y \leq Y_k \leq y + \delta) \approx \mathbf{P}(A)\mathbf{P}(B) = \lambda\delta \frac{\lambda^{k-1}y^{k-1}e^{-\lambda y}}{(k - 1)!},$$

from which we obtain

$$f_{Y_k}(y) = \frac{\lambda^k y^{k-1} e^{-\lambda y}}{(k - 1)!}.$$

- You are told that you are the 56th person in line. Callers depart according to a Poisson process with a rate of $\lambda = 2$ per minute. How long will you have to wait on the average, and what is the probability you will wait for more than 30 minutes?

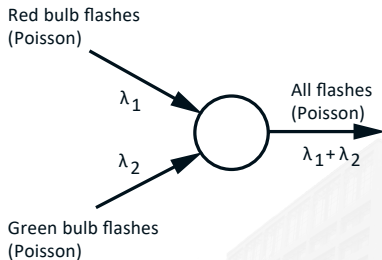
- You are told that you are the 56th person in line. Callers depart according to a Poisson process with a rate of $\lambda = 2$ per minute. How long will you have to wait on the average, and what is the probability you will wait for more than 30 minutes?
- The waiting time in minutes, defined as Y , is Erlang of order 56, and

$$\mathbf{E}[Y] = \frac{56}{\lambda} = 28.$$

- The probability that you have to wait for more than 30 minutes is given by

$$\mathbf{P}(Y \geq 30) = \int_{30}^{\infty} \frac{\lambda^{56} y^{55} e^{-\lambda y}}{55!} dy.$$

- Sum of independent Poisson **RVs** is Poisson
- Merging of independent Poisson **processes** is Poisson



- What is the probability that the next arrival comes from the first process?

$$\mathbf{P}(\text{Arrival is red} \mid 1 \text{ arrival}) \approx \frac{\lambda_1 \delta}{(\lambda_1 + \lambda_2) \delta}$$

- Let $N, X_1, X_2, \dots$ be independent RVs, where N takes nonnegative integer values.
- Let $Y = X_1 + \dots + X_N$ for positive values of N , and let $Y = 0$ when $N = 0$.

X_i	N	Y
Bernoulli, p	binomial, (m, q)	binomial, (m, pq)
Bernoulli, p	Poisson, λ	Poisson, λp
geometric, p	geometric, q	geometric, pq
exponential, λ	geometric, q	exponential, λq

- Recall the transform functions of the distributions:

$$M_{\text{Pois}}(s) = e^{\lambda(e^s - 1)}$$

$$M_{\text{Bern}}(s) = 1 - p + pe^s$$

$$M_{\text{Bino}}(s) = (1 - q + qe^s)^m$$

$$M_{\text{Expo}}(s) = \frac{\lambda}{\lambda - s}$$

$$M_{\text{Geom}}(s) = \frac{pe^s}{1 - (1 - p)e^s}$$

- Transform of sum of RVs, applying $M_Y(s) = M_N(\log M_X(s))$

$$M_{Y_1}(s) = (1 - q + q(1 - p + pe^s))^m = (1 - pq + pqe^s)^m$$

$$M_{Y_2}(s) = e^{\lambda((1-p+pe^s)-1)} = e^{\lambda p(e^s-1)}$$

- If X_i is Bernoulli with parameter p , and N is binomial with parameters m and q , then Y is binomial with parameters m and pq .

Hint:

- Split the Bernoulli process by keeping successes with probability q and discarding them with probability $1 - q$.
- Then Y represents the number of successes in the split process during the first m trials.
- Since the split process is Bernoulli with parameter pq , it follows that Y is binomial with parameters m and pq .

- If X_i is exponential with parameter λ , and N is geometric with parameters q , then Y is exponential with parameter λq .

Hint:

- View $X_1, X_1 + X_2, \dots$ as the arrival times in a Poisson process with parameter λ .
- Each arrival is rejected with probability $1 - q$ and is accepted with probability q , and N is the number of arrivals until the first acceptance.
- The process of accepted arrivals is obtained by splitting a Poisson process and is therefore itself Poisson with parameter λq .
- Y is the time of the first accepted arrival and is therefore exponential with parameter λq .

1. Review
2. The Bernoulli process
3. The Poisson process
4. **Recitation**
5. Summary



Let X , Y , and Z be independent exponential RVs with parameters λ , μ , and ν , respectively. Find $\mathbf{P}(X < Y < Z)$.



1. Review
2. The Bernoulli process
3. The Poisson process
4. Recitation
5. Summary



- Bernoulli process
 - Independence and memorylessness
 - Interarrival times and the k th arrival time
 - Splitting and merging
- Poisson process
 - Independence and memorylessness
 - Interarrival times and the k th arrival time
 - Merging
- Sum of random number of RVs
- Readings: Section 6.1-6.2
- Assignment: Problem Set 14
- Next lecture: Course review